

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)

CHARUSAT Campus, Changa – 388 421.

M Sc (BIOTECHNOLOGY/BIOCHEMISTRY/MICROBIOLOGY)

SEMESTER I EXAMINATION (BACKLOG)

BT702/MI702/BC702: CELL BIOLOGY

Date: 27.04.2011

Time: 10.00 a.m. to 1.00 p.m.

Total Marks: 70

Instructions to the candidates:

1. Provide all necessary information as required in the answer sheet.
2. Attempt all questions.
3. Questions in **Part A** should be answered in the question paper itself. Please **do not write** your candidate ID number on the question paper for Part A.
4. Write answers for Section I and Section II of **Part B** in separate answer sheets.
5. Use of cell phones is strictly prohibited.
6. Use of non-programmable calculators may be allowed if needed.

PART A

Total marks: 20

Q1. Choose the correct option and put $\sqrt$ mark in front of it:

1. The mechanism by which cell engulfs external fluid is called
(a) Exocytosis (b) Endocytosis (c) Phagocytosis (d) Pinocytosis
2. Which of the following techniques is useful in studying localization of transmembrane proteins?
(a) FRAP (b) Patch Clamp (c) Freeze fracture (d) Electron Spin Resonance
3. Which of the following in eukaryotes are said to be descendents of cyanobacteria?
(a) Chloroplasts (b) Mitochondria (c) Vacuoles (d) Golgi bodies
4. Most biomembranes are basically made up of
(a) Cholesterol (b) Glycolipids (c) Sphingolipids (d) Phospholipids
5. Which of the following ions are responsible for maintaining resting membrane potential?
(a) Cations (b) Anions (c) K^+ (d) Na^+
6. Which of the following is **NOT** present in animal cells?
(a) mitochondria (b) cell membrane (c) cell wall (d) cytoplasm
7. The protein complex at the centromere that holds two copies of the duplicated chromosome together is known as
(a) Cytochrome (b) Kinetochore (c) Centromeric protein (d) Histones
8. Which of the following is responsible for protein transport
(a) Vesicles (b) Vacuoles (c) Lysosomes (d) Peroxisomes
9. Hormone receptors
(a) have a high affinity for their natural ligand (b) are always linked to G proteins.
(c) are always located in the plasma membrane of their target cells (d) are lipids in nature

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10. Which of the following can serve as second messenger?
(a) cAMP (b) Ca^{++} (c) IP3 (d) All the above
11. Testosterone is most likely to bind to
(a) membrane ion channel (b) enzyme linked membrane receptor
(c) G-protein linked membrane receptor (d) cytoplasmic receptors
12. What event would activate a G protein?
(a) Hydrolysis of GTP to GDP (b) Phosphorylation of GDP to GTP
(c) Replacement of GDP with GTP (d) Phosphorylation of GTP to GDP
13. What's the order that the three stages of cell signaling go in?
(a) 1.Signal transduction, 2.Cellular response, 3.Signal reception
(b) 1.Cellular response, 2.Signal transduction, 3.Signal reception
(c) 1.Cellular response, 2.Signal reception, 3.Signal transduction
(d) 1.Signal reception, 2.Signal transduction, 3.Cellular response
14. The extracellular matrix can include the following molecular components:
(a) glycosaminoglycans (b) glycoproteins (c) laminin (d) all of the above
15. The following cleaves Scc1 to break cross link between sister chromatids
(a) Separase (b) DNase (c) CDK (d) cyclin
16. *cdc* mutants are used to study the cell cycle
(a) *Saccharomyces cerevisiae* (b) *Xenopus laevis*
(c) *Escherichia coli* (d) none of the above

Q2. State True or False

1. Transmembrane linker proteins and intracellular attachments proteins are the important components of occluding junctions. [True] / [False]
2. Ca^{+2} is required for tight junction integrity. [True] / [False]
3. Acetyl choline receptor selectively allows the entry of sodium ions due to high affinity. [True] / [False]
4. Synthesis of DNA and chromosome replication takes place during S phase of cell cycle. [True] / [False]

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Part B

Total Marks: 50

Section I

Q.1(A) Write a shortnote on **any two** of the following. (6)

- (a) Role of cholesterol in cell membrane
- (b) Mechanism of transport through $\text{Na}^+ \text{K}^+$ ATPase
- (c) Functions of ion-channels

Q.1(B) Explain the Patch Clamp Technique (4)

OR

Q.1(B) Explain Freeze Fracture Technique for studying transmembrane proteins (4)

Q.2(A) Describe the functions of **any two** of the following (6)

- (a) Lysosomes (b) Vacuoles (c) Nucleus (d) Cytoskeleton

Q.2(B) Draw the structure of mitochondria. Enlist the main functions of this organelle. (4)

OR

Q.2(B) Draw the structure of Chloroplast. Enlist the main functions of this organelle. (4)

Section II

Q.3(A) Write shortnote on **any two** of the following (4)

- (a) Gap Junction (b) Tight Junction (c) Adherens Junction

Q.3(B) How does controlled degradation of extra cellular matrix occur in mammalian system? (3)

OR

Q.3(B) What are proteoglycans? Give the role of hyaluronan in extra cellular matrix. (3)

Q.3(C) Signals don't always require membrane bound receptors. Explain with examples. (3)

Q.4(A) Explain the MAP Kinase pathway (4)

OR

Explain the signaling by G protein-coupled receptors (GPCR) (4)

Q.4(B) Explain plant two component signaling system (3)

Q.4(C) Write a note on **any one** of the following (3)

- (i) cAMP (ii) IP3

Q.5(A) Explain **any three** of the following in one or two sentences: (6)

- (i) cyclin (ii) restriction point (iii) mitogens (iv) cohesins (v) cell cycle

Q.5(B) Discuss the molecular mechanisms for regulating mitotic events during the cell cycle. (4)

OR

(B) Discuss *Saccharomyces cerevisiae* and *Xenopus laevis* as experimental systems used to identify and isolate Cell-cycle control proteins. (4)